



HANDLING E-MAIL

INTRODUCTION TO EMAIL

- An **email** (electronic mail) is a digital message sent over the internet from a sender to one or more recipients.
- It operates on a **store-and-forward** model, meaning the message is sent to a server (the "post office") where it waits until the recipient retrieves it.

Roles:

- **Sender** 🖋️: The user or system that creates and transmits the message via an **Email Client** (like Outlook or Gmail) to an **Outgoing Mail Server**.
- **Receiver** 📧: The intended recipient who uses an **Email Client** to connect to an **Incoming Mail Server** to retrieve and read the message.

COMPONENTS OF AN E-MAIL BODY

Subject: A concise line of text that summarizes the email's purpose. It's critical for the receiver to quickly grasp the content.

Body: The main content of the message. This can be sent as **plain text** or richly formatted using **HTML** (Hyper Text Markup Language).

Attachments: Files (like PDFs, images, or documents) that are encoded and sent along with the main message body.

MAILER SYSTEM (EMAIL DELIVERY SYSTEM)

- A **mailer system** (or email delivery system) is the entire infrastructure—including the software, hardware, and protocols—that enables the exchange of electronic messages.
- It's essentially the full set of processes and components required to handle an email from the moment you hit "Send" to the moment the recipient reads it.

COMPONENTS OF A MAILER SYSTEM

Technical Component	Post Office Analogy 	Simple Role
MUA (Mail User Agent)	You, the Sender 	The program (Gmail, Outlook) you use to write and read the email.
MSA (Mail Submission Agent)	The Post Office Counter 	The first server that takes your letter (email) and makes sure the address is formatted correctly before sending it out.
MTA (Mail Transfer Agent)	The Mail Truck/Sorting Office 	The routers and servers that move the email across the internet until it reaches the recipient's city.
MDA (Mail Delivery Agent)	The Final Mail Carrier 	The last server that drops the email into the recipient's personal inbox folder.

COMPONENTS OF A MAILER SYSTEM

Email System



E-Mail Protocols

- A protocol is about a standard method used at each end of a communication channel, in order to properly transmit information
- In order to deal with your email, you must use a mail client to access a mail server.
- the most commonly used Email protocols on the internet - **POP3, IMAP and SMTP.**
- Each one of them has specific function and way of work.

SMTP (Simple Mail Transfer Protocol)

- **Role: Sending** mail. It's used by the **Mailer (MUA)** to push mail to the **Mail Server (MSA/MTA)**, and by **Mail Servers (MTAs)** to transfer mail between each other
- It was first proposed in 1982. It is a standard protocol used for sending e-mail efficiently and reliably over the internet.
- SMTP is application-level protocol.
- SMTP is connection-oriented protocol.
- SMTP is text-based protocol.
- It handles exchange of messages between e-mail servers over TCP/IP network.
- Simple Mail Transfer Protocol (SMTP) is a TCP/IP protocol used in sending and receiving e-mail.
- **Analogy:** The process of **depositing your letter** at the counter and the **mail truck routes** across cities.

IMAP(Internet Message Access Protocol)

- **Role: Retrieving** mail. It allows the Mailer (MUA) to access and manage messages directly on the **Mail Server**.
- **Key Feature:** Emails **stay on the server**. This allows you to access the same inbox, folders, and message status (read/unread) from multiple devices.
- **Analogy:** Checking your P.O. Box directly—the letters remain there until you move them.

How IMAP Works (Simple Flow)

- The sender sends an email (via SMTP).
- The email is stored on the recipient's **mail server**.
- The recipient's device (phone, laptop, webmail) connects to the server using **IMAP**.
- Emails are displayed to the user **without needing to download them permanently**.
- Any actions (read, delete, move) are **updated on the server**, so all devices stay in sync.

POP3(Post Office Protocol Version3)

- **Role: Retrieving** mail
- **Key Feature:** Typically **downloads** the message to your local device and then deletes it from the server (though modern implementations often allow you to leave a copy)
- **Analogy:** An old-fashioned mail carrier delivers the letter, and you take it out of your mailbox.

How POP3 Works

- The sender's email arrives on the recipient's **mail server**.
- The recipient connects to the server using **POP3**.
- Emails are **downloaded** to the local device.
- (Optional) Emails are **removed from the server** after download.

SMTP Settings for Python

Setting	Description	Example (Gmail)
SMTP server	Mail server address	smtp.gmail.com
Port	Server port	587 (TLS) or 465 (SSL)
Sender email	Your email ID	example@gmail.com
Password / App Password	Used for authentication	your-app-password
Security type	TLS or SSL	TLS (587) recommended



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